



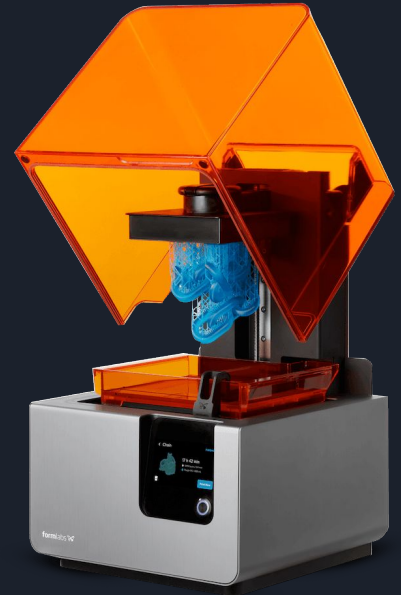
SLA 3D Printing

Parameters Effect on Surface Quality and Dimensions

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10MAR2025

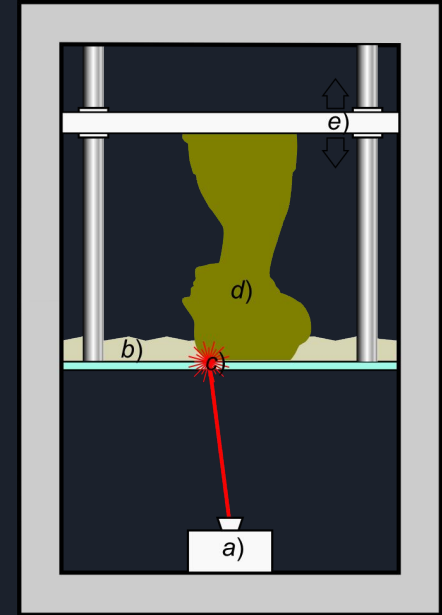
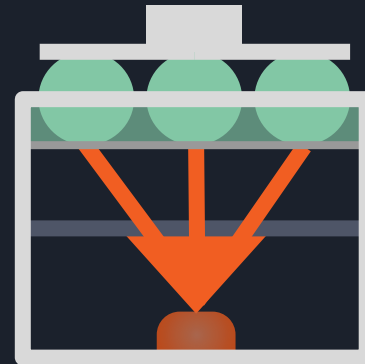
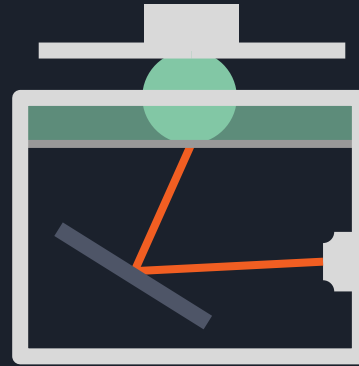
SLA 3D Printing Overview

- Stereolithography (SLA) the first commercialized 3D printing process
- Developed in the 1980s (patented in 1986)
- Works by focusing a UV laser into a vat of photopolymer resin
- Liquid material is called “resin”
- Benefits:
 - Print speed
 - High surface quality
 - Parts can be machined
- Disadvantages
 - Often costlier than other methods
 - Parts must be washed and cured before use
 - Resins can be messy



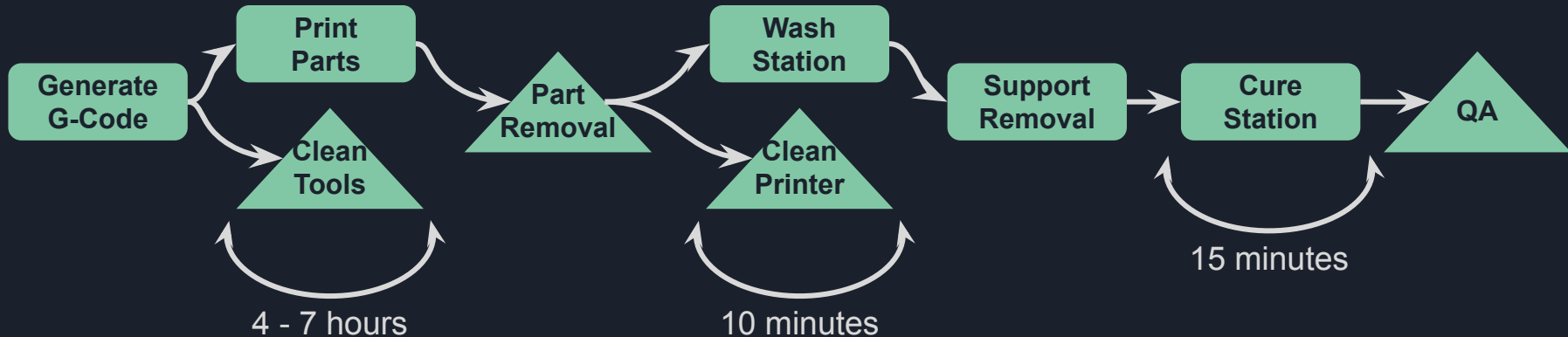
SLA 3D Printing Overview

- a) A laser selectively illuminates the transparent bottom (c)
- b) Tank filled with liquid photo-polymerizing resin
- d) Solidified resin (the desired part geometry)
- e) Lifting platform moves the solidified resin up as the part prints



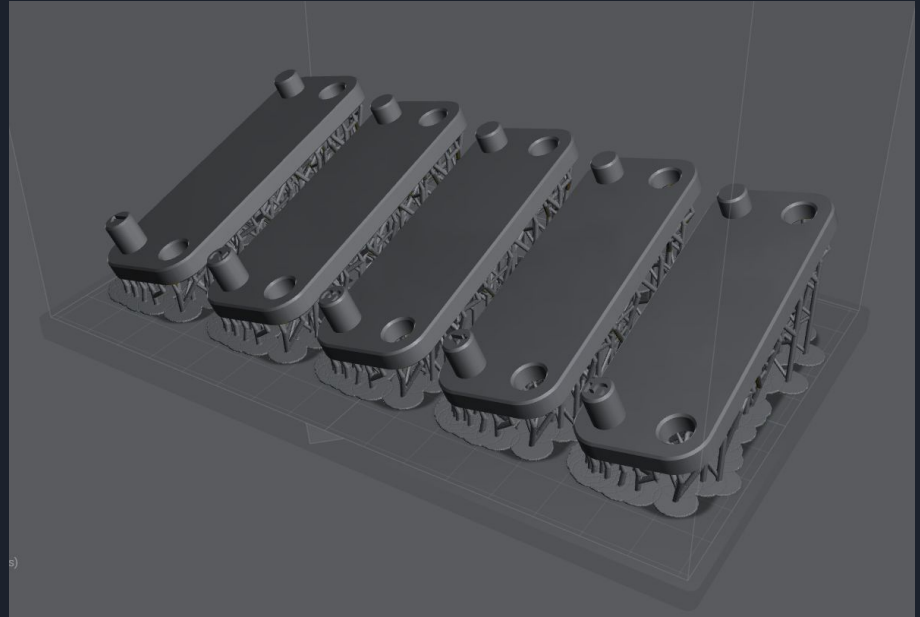
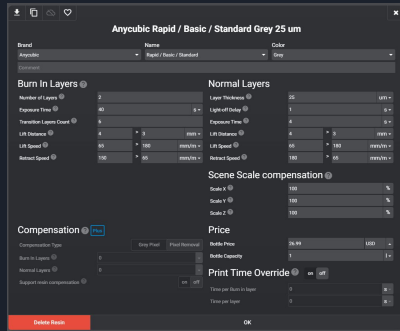
Manufacturing Process

- G-Code was generated using Lychee Slicer free edition
- Samples were printed on the Anycubic Photon M3
- Post processing utilized the Anycubic Wash and Cure station

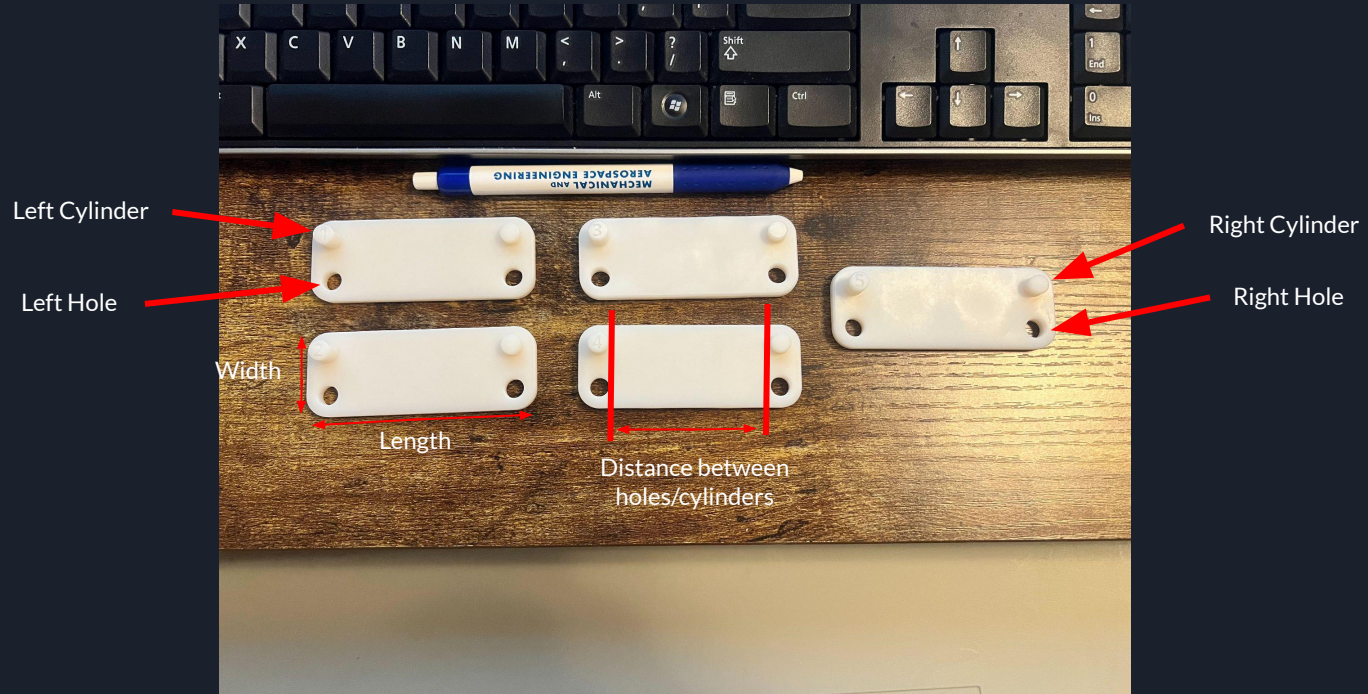


Print Parameters

- All prints were printed with the same base settings and the same supports
- Layer thickness varied by 5µm from 25µm to 45µm
- Exposure Time varied from 2 seconds to 4 seconds



Sample Parts



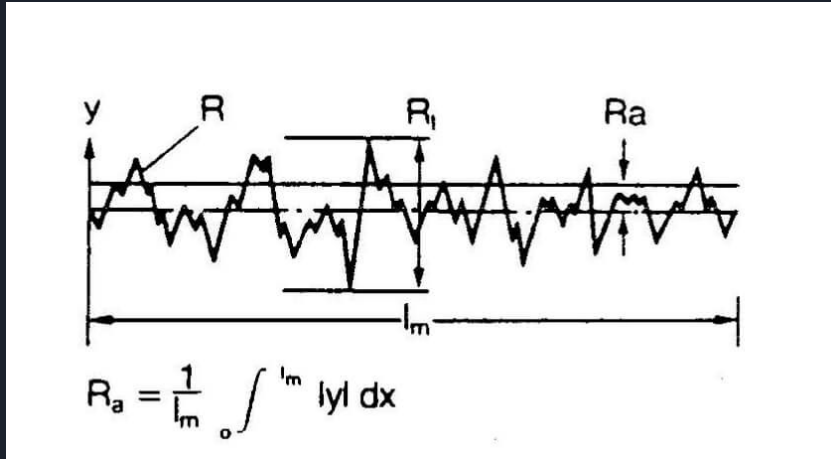
Inspection Methods

- Calipers
- Mahr MarSurf PS10
- Cylindrical Metric Gage Pins



Surface Roughness, Ra

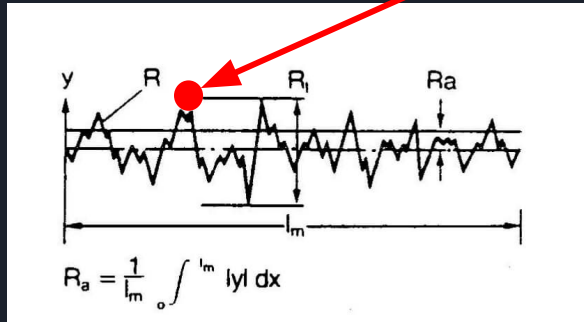
- Mahr MarSurf PS10 used for surface measurement
- Ra, arithmetic average surface roughness
- Average of absolute values of surface profile height deviation from the mean



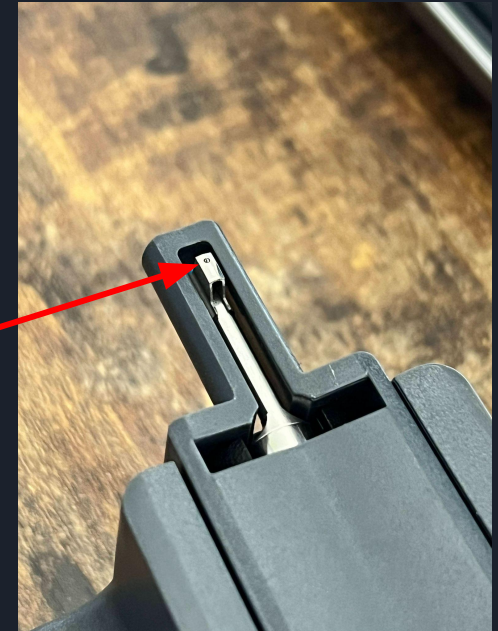
Surface Roughness Measurement

Mahr MarSurf PS10

- Traversing length (L_t): 15mm
- Measuring force (N): 0.00075
- Probe tip radius (μm): 2
- Measuring speed (mm/s): 0.5 to 1.0 mm/s



2 μm probe tip



Gage Pins - Hole Size

- Gage pins were used to measure the hole size of both the left and right holes
- The largest pin that will fit is the hole size



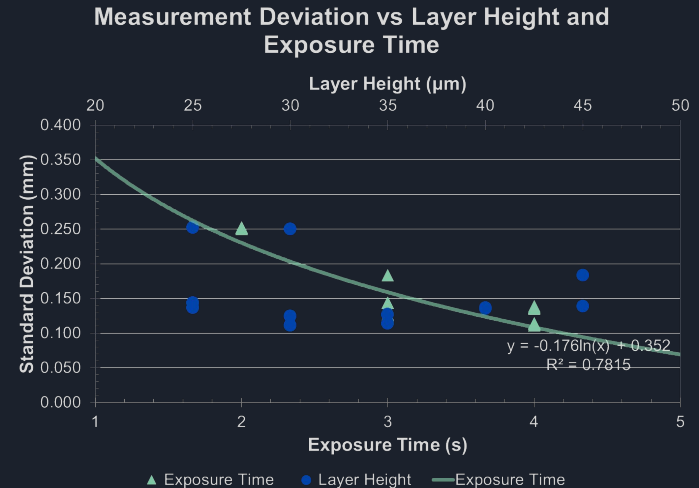
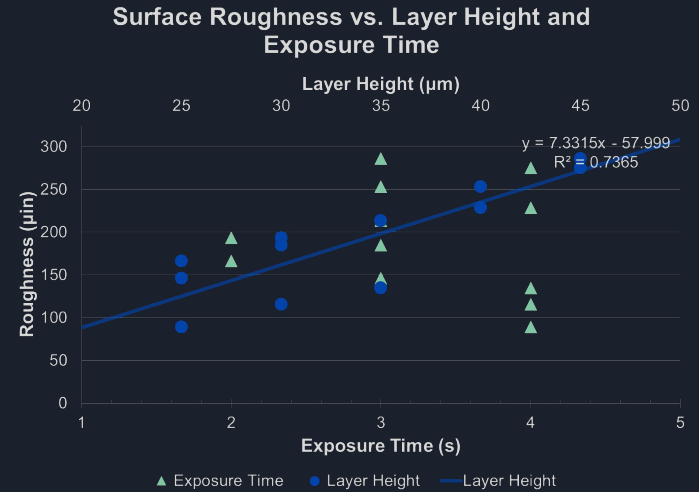
Overall Dimensions - Calipers

- Calipers were used to measure:
 - Length
 - Width
 - Minimum distance between holes
 - Minimum distance between cylinders
 - Cylinder diameter



Findings

- Surface roughness seems to have a positive correlation with layer height
 - Exposure time shows high scatter, with no clear relationship to surface finish
- Exposure time appears to have a negative correlation with measurement deviation
 - Layer height shows little to no effect on measurement deviation

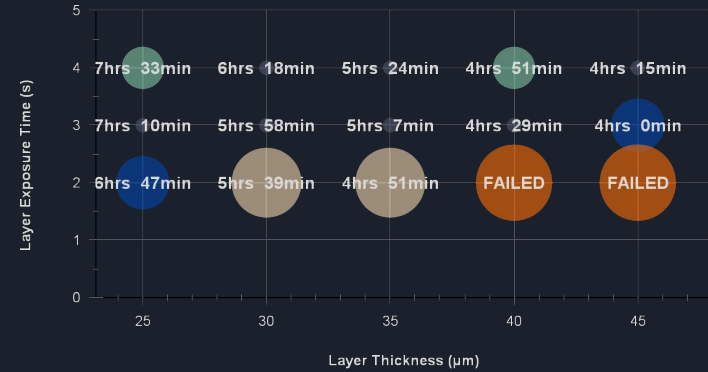


Failure Analysis Data

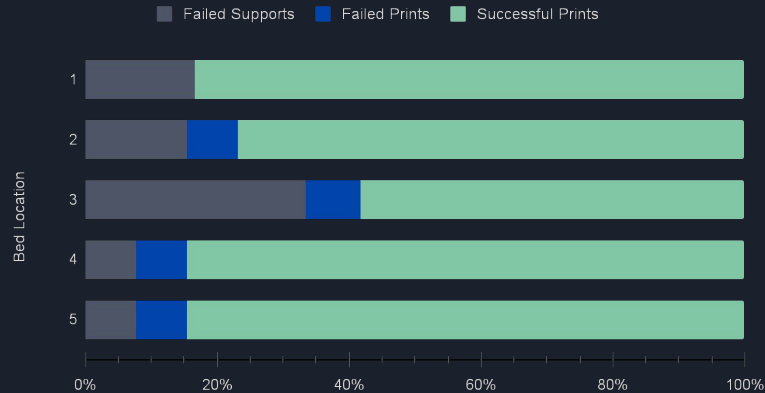
How Parameters and Environment Affect Part Failure Rate

- No strong correlation to ambient temperature
- Weak correlation to layer height
- Strong correlation to layer exposure time
- Correlated to bed location #3

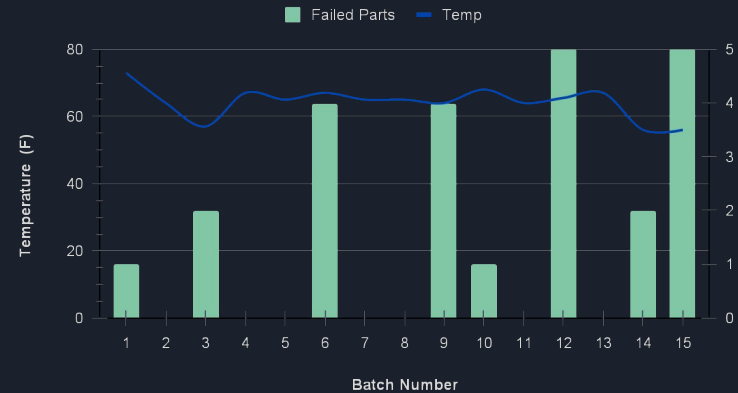
Effect of Layer Height and Exposure on Failure Rate



Failure Bed Locations



Effect of Temp on Number of Failed Parts

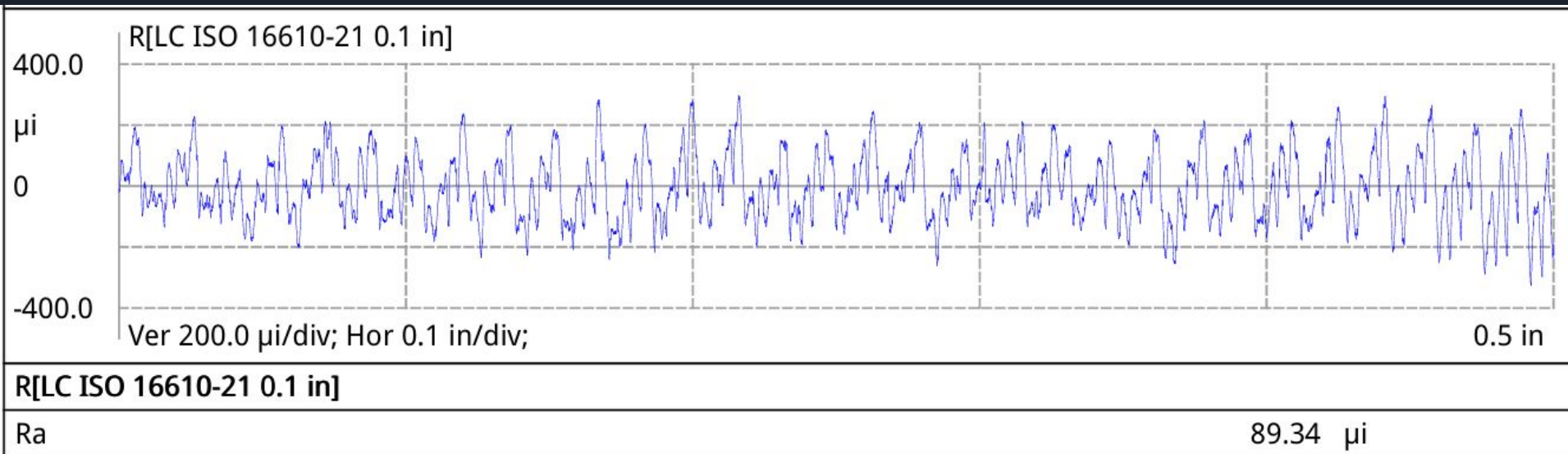




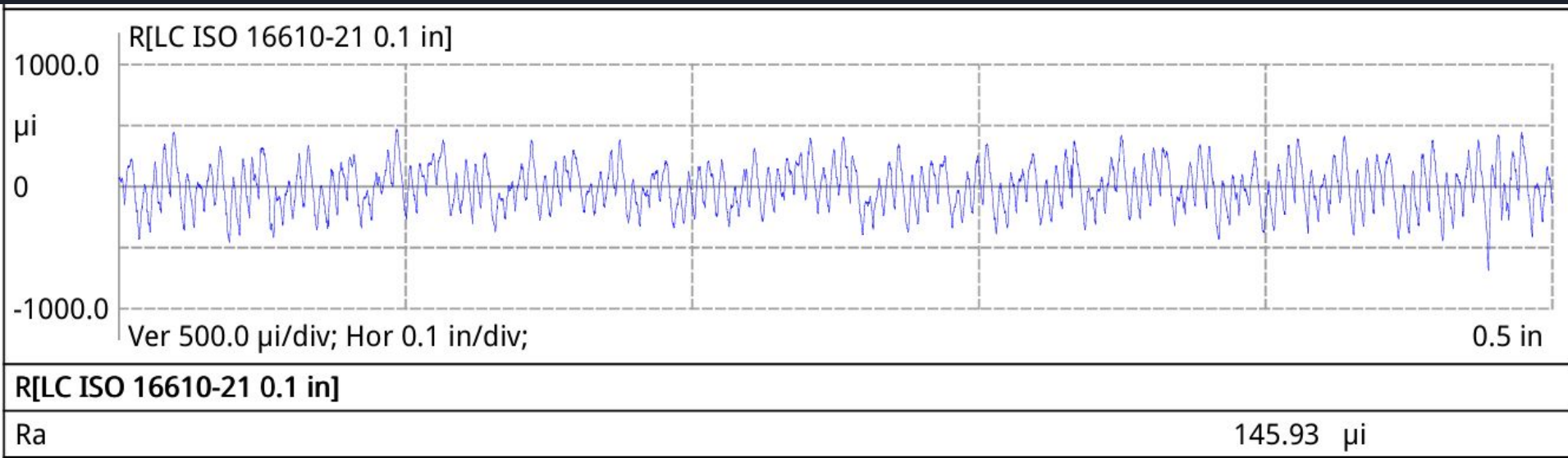
Conclusions

- Resulting Recommendations:
 - Surface roughness can be controlled by adjusting layer thickness
 - Measurement error increases with lower layer exposure time
 - Lower layer heights showed more inconsistent measurement data
 - If surface finish is not critical and feature size is sufficiently large, a layer height of 45 μm with 4 second layer exposure showed the least print time with low chance of failed prints
- Future study:
 - Exposure times between 2 to 3 seconds, which seems to be the limit around where prints fail
 - Study anti-aliasing mode with different layer thickness and exposure time
 - Print multiple batches of each layer height and exposure time to better determine failure rates

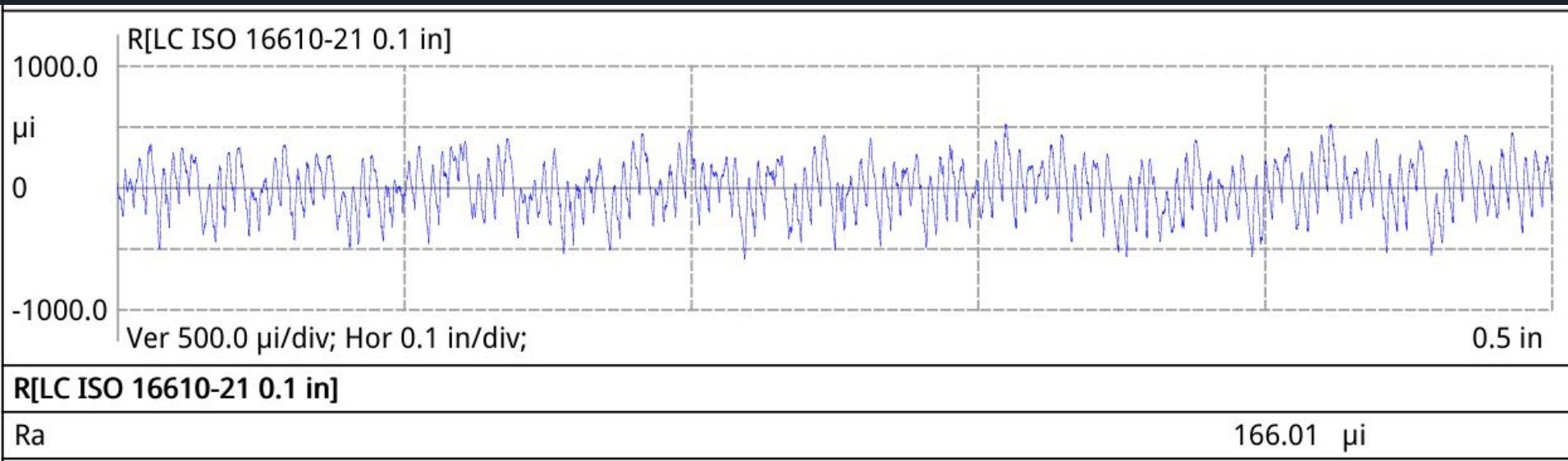
Appendix - Batch 1-1 Part 1



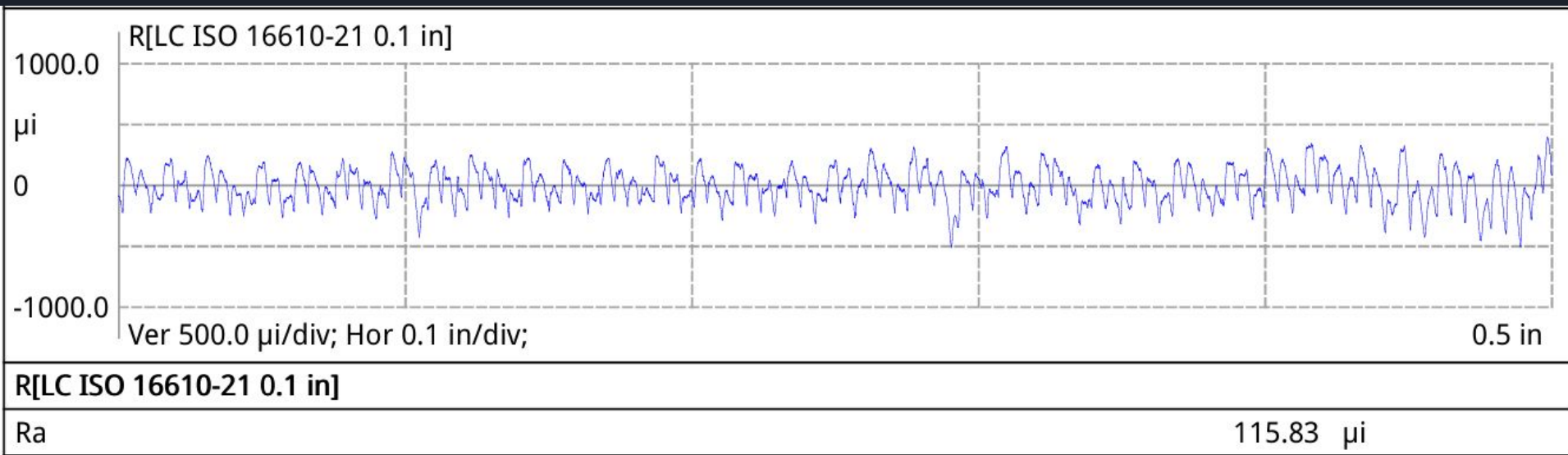
Appendix - Batch 1-2 Part 1



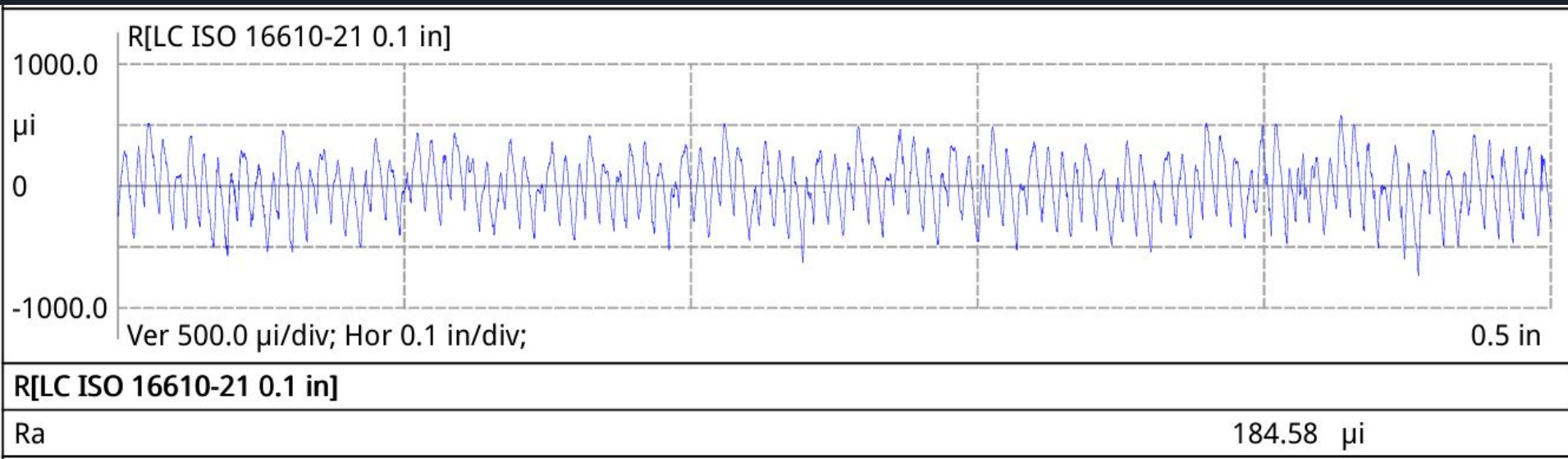
Appendix - Batch 1-3 Part 1



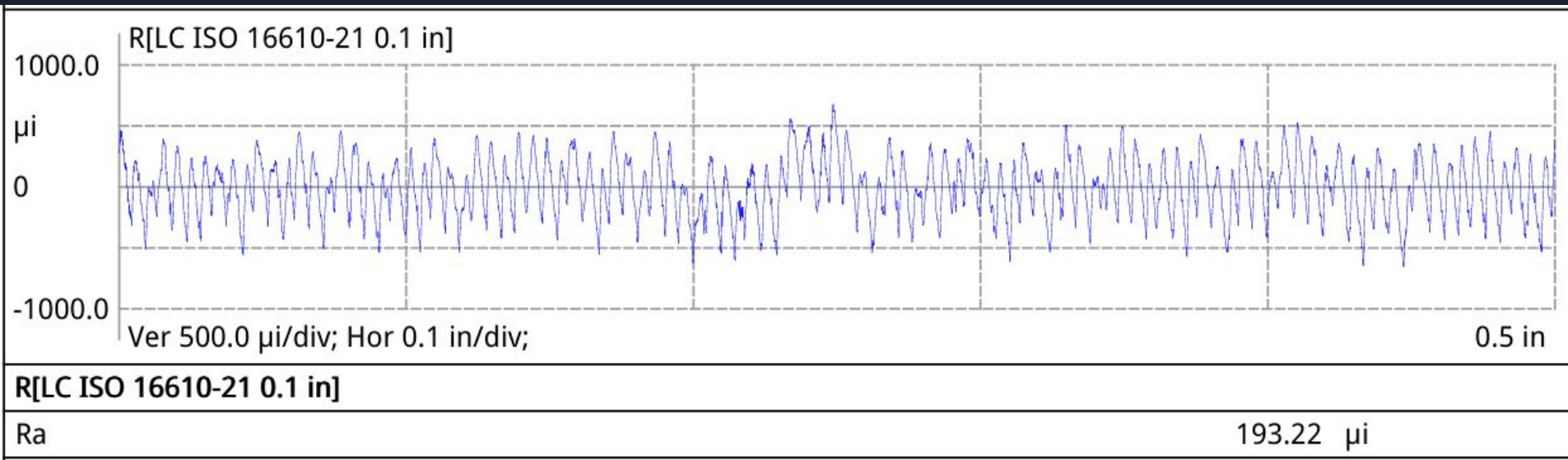
Appendix - Batch 2-1 Part 1



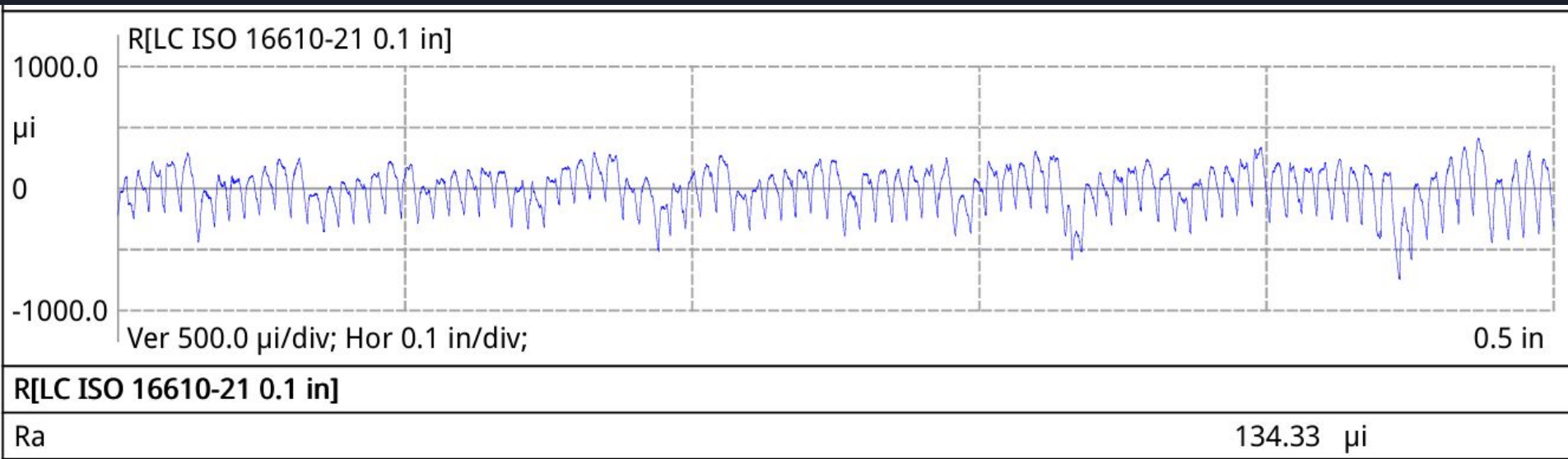
Appendix - Batch 2-2 Part 1



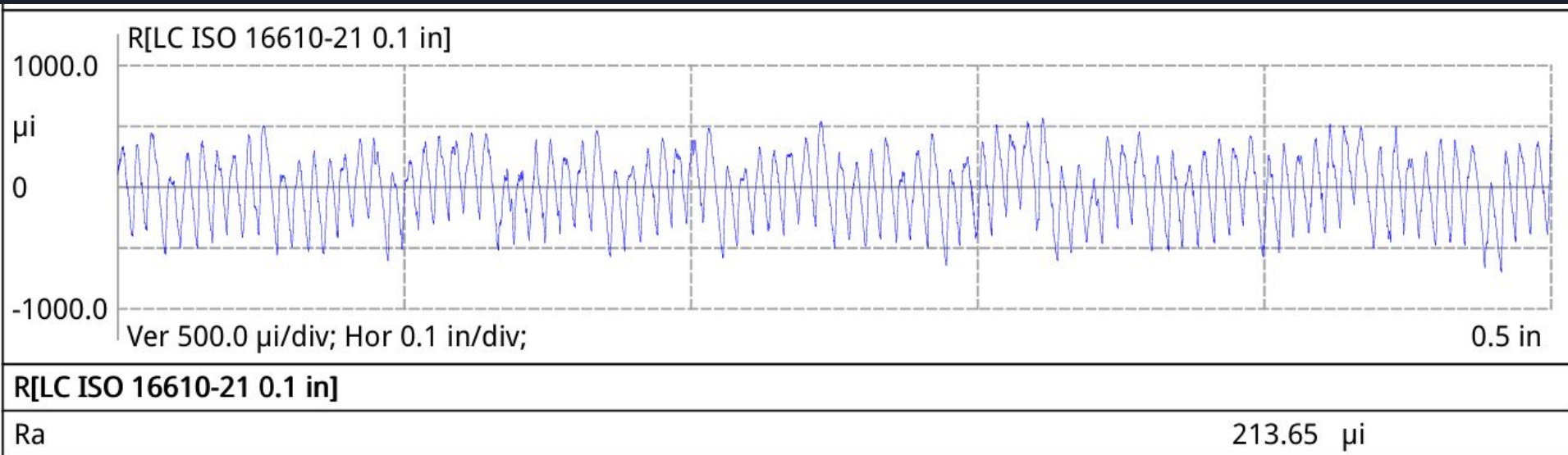
Appendix - Batch 2-3 Part 2



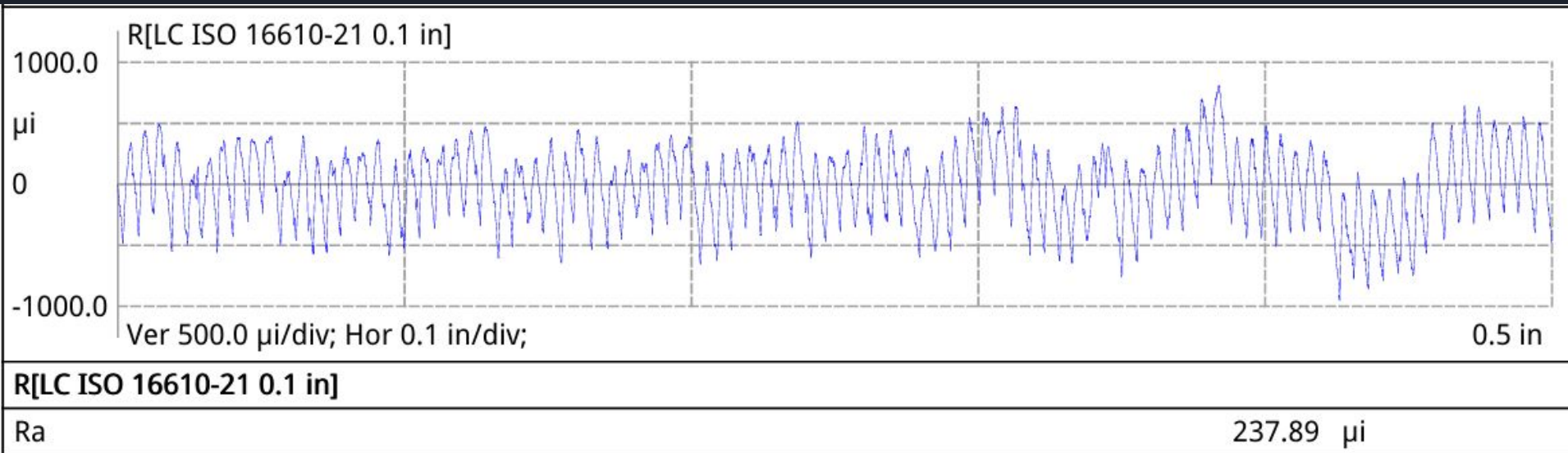
Appendix - Batch 3-1 Part 1



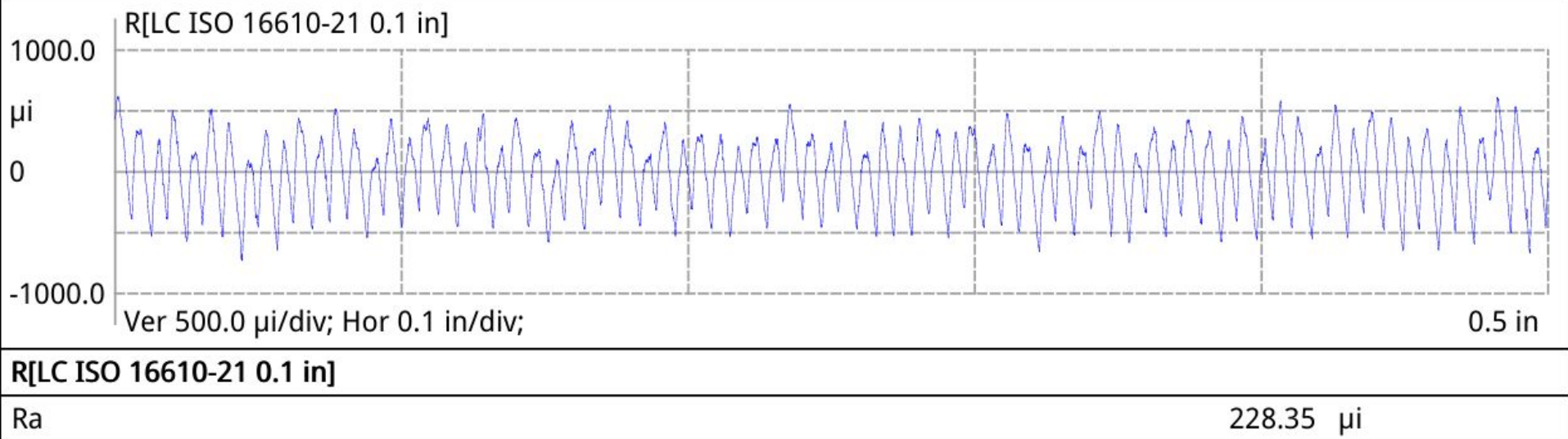
Appendix - Batch 3-2 Part 1



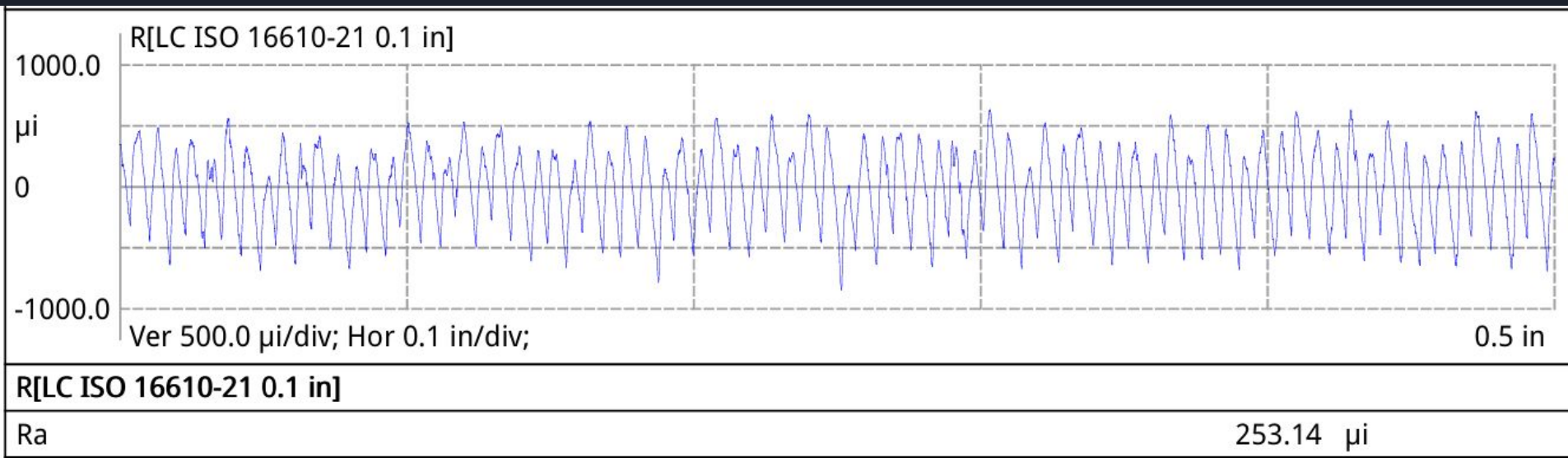
Appendix - Batch 3-3 Part 1



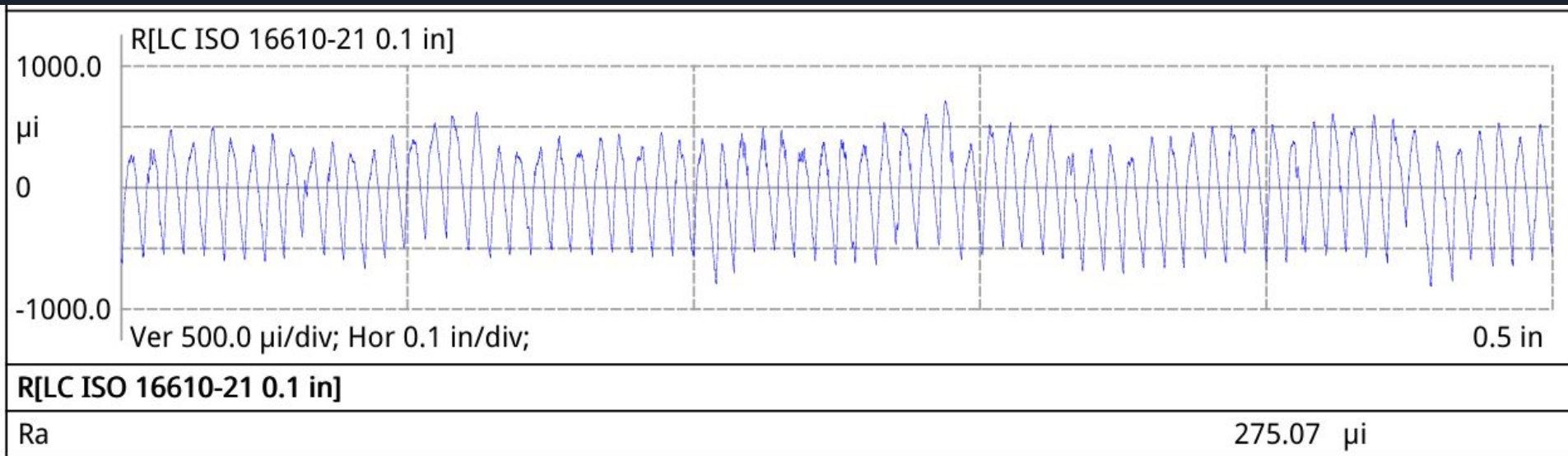
Appendix - Batch 4-1 Part 1



Appendix - Batch 4-2 Part 1



Appendix - Batch 5-1 Part 1



Appendix - Batch 5-2 Part 1

